

Design, Implementation and Simulation of an 8-Point FFT Circuit in 45nm CMOS

¹Uppal Das, ²Promit Chakraborty

Department of Electrical Engineering

Indian Institute of Science, Bengaluru

Bengaluru, India

¹uppaldas@iisc.ac.in, ²promitc@iisc.ac.in

Abstract—The 8-Point FFT Circuit, implemented in 45nm CMOS, represents an optimized digital circuitry designed for high-speed computation, energy efficiency, and cost-effectiveness in terms of area and power. The development cycle involves the design, testing and simulation through Icarus Verilog, synthesis employing Yosys with the Nangate Open Cell 45nm standard cell library, and subsequent analysis conducted in terms of timing and power using OpenSTA. This comprehensive approach ensures the robustness and efficiency of the circuit, aligning with contemporary standards and addressing the critical considerations of performance, energy consumption, and design economy.

Index Terms—Fast Fourier Transform, Carry Select Adder, Borrow Select Subtractor, Fixed Point Arithmetic

I. INTRODUCTION

Fast Fourier Transform (FFT) is an algorithm employed to compute the Discrete Fourier Transform (DFT) or its inverse (IDFT) for a given sequence. In this particular design, each input element, denoted as in_m (where $m = 0,1,...,7$), is a complex number. The real and imaginary components of each input are represented as signed 16-bit fixed-point quantities, encompassing 1 sign bit, 7 bits for the decimal part, and 8 bits for the fractional part. Similarly, each output element, designated as out_m (where $m = 0,1,...,7$), is a complex number. The real and imaginary parts of the output are also represented as signed 16-bit fixed-point quantities, featuring 1 sign bit, 7 bits for the decimal part, and 8 bits for the fractional part. In the context of designing an efficient 8-point FFT circuit utilizing a 45nm CMOS process, the primary objective revolves around striking a balance between key trade-offs, namely speed, power consumption, area utilization, and accuracy.

II. ARCHITECTURE

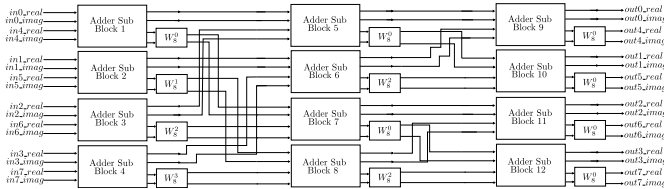


Fig. 1: Architecture of 8-Point FFT Circuit

The 8-Point FFT circuit comprises 12 adder-subtractor blocks and 12 twiddle factor blocks (W_8^0 (7 no.), W_8^1 (1 no.), W_8^2 (3 no.), and W_8^3 (1 no.)), illustrated in Figure 1.

The fundamental components of this circuit are basic arithmetic circuits, namely adders, subtractors, and multipliers. The following provides descriptions for each block depicted in Figure 1:

A. Adder-Subtractor Block

The adder-subtractor block is composed of two adders and two subtractors, each equipped to perform 16-bit signed addition and 16-bit signed subtraction, respectively.

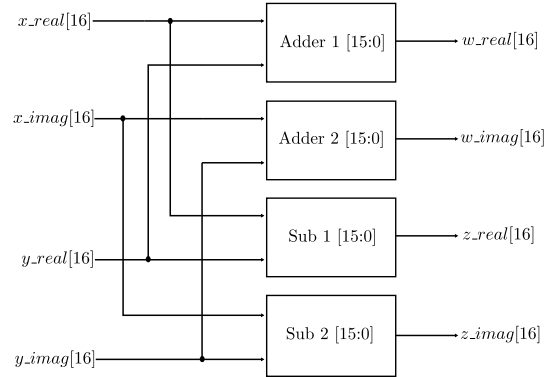


Fig. 2: Internal Structure of Adder Subtractor Block

In Figure 2, the variables x_real and x_imag signify the real and imaginary components of one input respectively, whereas y_real and y_imag denote the real and imaginary components of another input respectively. Similarly, w_real and w_imag represent the real and imaginary components of one output respectively, while z_real and z_imag correspond to the real and imaginary components of another output respectively.

B. Twiddle Factor Block

The twiddle factor blocks (W_8^0 , W_8^1 , W_8^2 , W_8^3) for the 8-point FFT are visually represented in Figure 3. Within the context of the 8-Point FFT, the W_8^0 block operates as a buffer, eliminating the necessity for additional digital circuitry. The W_8^2 block functions as a cross buffer, incorporating a multiplication factor of -1 to the real part of the input. The W_8^1

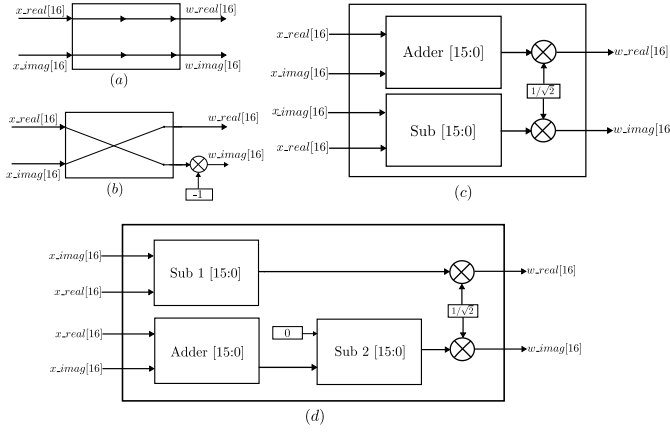


Fig. 3: (a) W_8^0 Block, (b) W_8^2 Block, (c) W_8^1 Block, (d) W_8^3 Block

and W_8^3 blocks involve multiplication by $\frac{1}{\sqrt{2}}$ and incorporate a sequence of additions and subtractions, as illustrated in Fig-3(c) and 3(d). Within this framework, the adder and subtractor perform 16-bit signed addition and subtraction, respectively. The architectural design implemented for both the adder and subtractor aligns with that of the adder-subtractor block. In Figure 3, x_real and x_imag denote the real and imaginary components of the input, while w_real and w_imag represent the real and imaginary components of the output.

C. Adder Selection

For the addition of 16-bit signed integers, the selected adder is the Carry-Select Adder (CSA). This choice is motivated by its advantages over alternative adder architectures, particularly in terms of speed and parallelism.

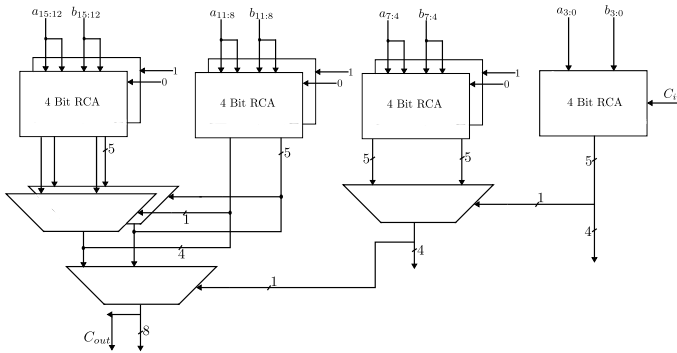


Fig. 4: Signed Carry Select Adder

The CSA architecture partitions the input into multiple blocks, each operating independently. This segmentation results in a reduced critical path delay when compared to serial adders. The subdivision of the adder into smaller blocks serves to restrict carry propagation within each block, thereby contributing to an overall decrease in propagation delay.

D. Subtractor Selection

The Borrow Select Subtractor (BSS) has been selected for the purpose of 16-bit signed subtraction. This choice

is rooted in its ability to accurately handle signed integers, incorporating consideration of the signed bit and employing the 2's complement representation as needed.

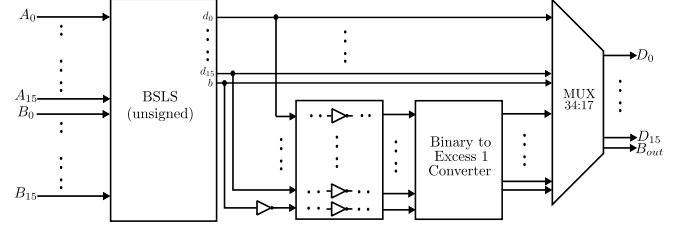


Fig. 5: Signed Borrow Select Subtractor

This methodology guarantees accurate results for both positive and negative numbers. In situations where overflow or underflow conditions may potentially occur, the Borrow-Select Subtractor adeptly mitigates such concerns by dynamically adjusting the subtraction method based on the comparison of the operands.

E. Multiplier Selection

For the multiplication of two 16-bit signed registers, with the necessity to store the result in a 16-bit register, recourse must be taken to fixed-point arithmetic.

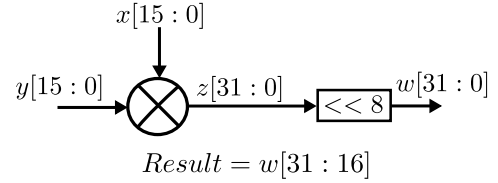


Fig. 6: Signed Fixed Point Multiplier

Initially, the multiplication process involves multiplying two 16-bit registers, and the resultant product is stored in a 32-bit register. Subsequently, a left shift by 8 bits is executed, and within the outcome, the most significant 16 bits are designated for the storage. In this manner, the result can be accommodated in a 16-bit register in the 8.8 fixed-point format.

III. VERILOG HARDWARE DESCRIPTION

Figures 1, 2, and 3 illustrate the structure of the implemented digital circuits. Analysis of Figure 3 reveals that the last two stages exclusively necessitate addition and subtraction operations, for which Carry Select Adder and Borrow Select Subtractor has been chosen, as previously discussed. In particular, the multiplication operation with $\frac{1}{\sqrt{2}}$ is solely applicable in the first stage, involving two twiddle factors. The initial cycle encompasses the operations of the first stage adder subtractor block and the twiddle factor stage block. Subsequently, the second i.e the last cycle is dedicated to the second and third stage adder-subtractor blocks.

IV. PERFORMANCE

The performance evaluation encompasses synthesizing the design utilizing Yosys with the Nangate Open Cell 45nm standard cell library in the typical, slow and fast TT corner. Concurrently, the timing characteristics and power consumption of the implementation is meticulously assessed through OpenSTA.

A. Process Corner

The digital circuit undergoes testing in three distinct Process-Voltage-Temperature (PVT) corners, namely fast, typical, and slow. The definitions for each of these PVT corners are as follows:

Process	Voltage	Temperature
Fast	1.1V	-40 °C
Typical	1.0V	27 °C
Slow	0.9V	125 °C

B. Clock Cycle per FFT Computation

The designed digital circuit necessitates 2 clock cycles after the arrival of input for the computation of the Fast Fourier Transform (FFT) of the input.

C. Synthesis using Yosys

The design was synthesized in Yosys using 45 nm technology. In Fig-7, the upper-level module, denoted as 'Eight-PointFFT,' comprises 24 instances of 'BorrowSelectSub,' 34 instances of 'CarrySelectAdd,' and 4 instances of 'FixedPointMultiplier' modules.

```

=== design hierarchy ===
EightPointFFT      1
BorrowSelectSub    24
CarrySelectAdd     34
FixedPointMultiplier 4

Number of wires:      18909
Number of wire bits:  18909
Number of public wires: 9173
Number of public wire bits: 9173
Number of memories:   0
Number of memory bits: 0
Number of processes:  0
Number of cells:      10762
AND2_X1             544
AND3_X1             360
AND4_X1             68
AOI211_X1           28
AOI21_X1            870
AOI221_X1           28
AOI22_X1            198
DFF_X1              576
INW_X1              1036
NAND2_X1            1280
NAND3_X1            168
NAND4_X1            32
NOR2_X1             1304
NOR3_X1             180
NOR4_X1             4
OAI211_X1           50
OAI21_X1            840
OAI22_X1            24
OAI33_X1            42
OR2_X1              176
OR3_X1              362
OR4_X1              4
XNOR2_X1            1894
XOR2_X1             774

Chp area for top module '\EightPointFFT': 13991.600000

Warnings: 2132 unique messages, 2196 total
End of script. Logfile hash: ab36e0232
CPU: user 11.12s system 0.12s, MEM: 40.73 MB total, 34.64 MB resident
Yosys 0.9 (git sha1 1979e0b)
Time spent: 32% 2x write_verilog (3 sec), 15% 14x opt_merge (1 sec), ...

```

Fig. 7: Yosys Synthesis Report for Design

The total area of synthesized design is $13991.6 \mu m^2$.

D. Static Timing Analysis using OpenSTA

The Static Timing Analysis (STA) was conducted across all process corners, including fast, typical, and slow conditions. The STA report, illustrated in Fig-8, encompasses the analysis results for fast, typical, and slow scenarios.

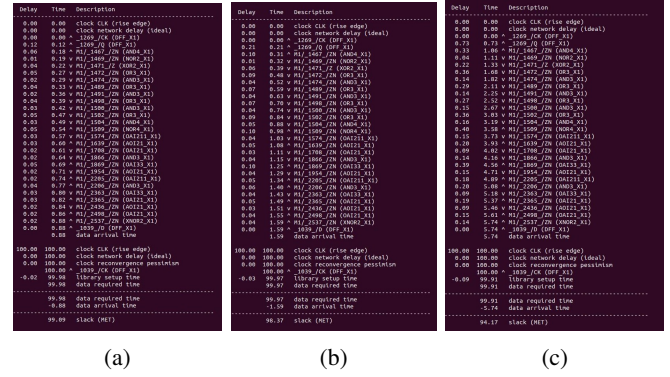


Fig. 8: STA Report at (a) Fast, (b) Typical and (c) Slow Process Corners

A clock time period of 100ns was utilized in the analysis. The calculated slack values for each process corner are presented in the table below.

Process	Slack (in nS)
Fast	99.09
Typical	98.37
Slow	94.17

For the circuit to operate reliably under all conditions, it is imperative that the minimum clock speed satisfies the requirement $T_{CLK} \geq 100 - 94.17 = 5.83ns$. Consequently, to ensure functionality across all scenarios, the clock frequency should be constrained to values less than or equal to 0.1715 GHz.

E. Power Consumption

The power consumption was quantified through the utilization of OpenSTA. Fig-9 depicts the simulation results delineating power consumption across various process corners.

Process	Internal(mW)	Switching(mW)	Leakage(mW)	Total(mW)
Fast	0.467	0.13	1.04	1.63
Typical	0.331	0.1	0.308	0.739
Slow	0.241	0.0732	0.198	0.511

The power consumption of the circuit at maximum operating frequency (0.1715 GHz) is tabulated in the above table.

F. Energy Consumption

The power consumption at different process corners was observed using OpenSTA. Additionally, utilizing GTK-Wave, we determined that it requires two clock cycles to compute the FFT of the input.

Process	Internal(pJ)	Switching(pJ)	Leakage(pJ)	Total(pJ)
Fast	5.44	1.514	12.11	18.98
Typical	3.86	1.166	3.59	8.618
Slow	2.81	0.8535	2.308	5.958

By considering the power consumption data and the maximum operating frequency (0.1715 GHz), we can deduce the energy consumption throughout the entire FFT computation. Different energy consumption at different process corners has been tabulated in the above table.

OpenSTA> report_power					
Group	Internal Power	Switching Power	Leakage Power	Total Power	
Sequential	1.32e-04	2.09e-05	1.59e-04	3.13e-04	19.1%
Combinational	3.34e-04	1.09e-04	8.78e-04	1.32e-03	80.9%
Macro	0.00e+00	0.00e+00	0.00e+00	0.00e+00	0.0%
Pad	0.00e+00	0.00e+00	0.00e+00	0.00e+00	0.0%
Total	4.67e-04	1.30e-04	1.04e-03	1.63e-03	100.0%
	28.6%	8.0%	63.5%		

(a)

OpenSTA> report_power					
Group	Internal Power	Switching Power	Leakage Power	Total Power	
Sequential	9.56e-05	1.61e-05	4.56e-05	1.57e-04	21.3%
Combinational	2.36e-04	8.39e-05	2.62e-04	5.82e-04	78.7%
Macro	0.00e+00	0.00e+00	0.00e+00	0.00e+00	0.0%
Pad	0.00e+00	0.00e+00	0.00e+00	0.00e+00	0.0%
Total	3.31e-04	1.00e-04	3.08e-04	7.39e-04	100.0%
	44.8%	13.5%	41.6%		

(b)

OpenSTA> report_power					
Group	Internal Power	Switching Power	Leakage Power	Total Power	
Sequential	7.02e-05	1.19e-05	3.14e-05	1.13e-04	22.2%
Combinational	1.70e-04	6.14e-05	1.66e-04	3.98e-04	77.8%
Macro	0.00e+00	0.00e+00	0.00e+00	0.00e+00	0.0%
Pad	0.00e+00	0.00e+00	0.00e+00	0.00e+00	0.0%
Total	2.41e-04	7.32e-05	1.98e-04	5.11e-04	100.0%
	47.0%	14.3%	38.6%		

(c)

Fig. 9: Power Consumption Report at (a) Fast, (b) Typical and (c) Slow Process Corners

V. RESULTS

The design underwent simulation in GTKWave, and the resulting waveforms are visually represented in Fig-10. It is discernible from the waveforms that the ultimate stable output is achieved after the arrival of the input signal over a duration of 2 clock cycles.

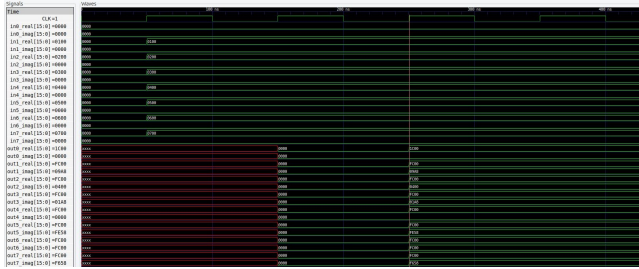


Fig. 10: Output Waveforms from GTKWave

The table below presents values obtained from simulation in 8.8 Fixed Point Format. Both the real and imaginary components of all eight inputs are displayed.

Index	Real (in Hex)	Value	Imag (in Hex)	Value
0	1C00	28	0000	0
1	FC00	-4	09A8	9.656
2	FC00	-4	0400	4
3	FC00	-4	01A8	1.656
4	FC00	-4	0000	0
5	FC00	-4	FE58	-1.344
6	FC00	-4	FC00	-4
7	FC00	-4	F658	-10.344

The subsequent table displays the absolute error between results derived from analytical calculations and those obtained through simulations.

X(k)	Expected	Obtained	Error
X(0)	28	28	0
X(1)	-4 + 9.656j	-4 + 9.656j	0
X(2)	-4 + 4j	-4 + 4j	0
X(3)	-4 + 1.656j	-4 + 1.656j	0
X(4)	-4	-4	0
X(5)	-4 - 1.656j	-4 - 1.344j	0.312j
X(6)	-4 - 4j	-4 - 4j	0
X(7)	-4 - 9.656j	-4 - 10.344j	0.688j

Upon examination of the above table, it is observed that the majority of outputs exhibit zero error, with the exception of two instances. The errors in these specific outputs stem from the Fixed Point Multiplication. Using a complex fixed-point multiplier with complex algorithms could have mitigated these errors; however, such an enhancement would inevitably lead to increased area, power and energy consumption.

VI. WAYS TO CALCULATE INVERSE 8 POINT FFT

The code used for the calculation of the 8-point Fast Fourier Transform (FFT) can be used to compute the Inverse FFT by interchanging the first and last stages. Additionally, it is essential to provide the conjugates of the twiddle factors in this modified configuration.

VII. ARCHITECTURAL TRADEOFFS AND OTHER DESIGN TECHNIQUES

In the context of this design, a marginal sacrifice in accuracy has been made to optimize for reduced area, power, and energy consumption. Carry-select adders and borrow-select subtractors have been employed in our circuit design, contributing to increased operational speed and, consequently, reduced power consumption and energy utilization. Specifically, in the case of an 8-point Fast Fourier Transform (FFT) circuit, a simple fixed-point multiplier has been chosen for implementation. Since the 8 point FFT circuit requires four signed multiplications, and opting for multipliers with intricate algorithms has been consciously avoided in favor of simplicity and efficiency. While a more complex multiplier circuit could be considered for higher-order FFT circuits.

CONTRIBUTION

- Uppal Das has done the coding of the fundamental arithmetic circuits, the initial adder-subtractor block, and the twiddle factor block and also has done the synthesis for the typical process corner and has created the schematics utilized in this report.
- Promit Chakraborty has done the coding for the last two stages, along with the synthesis for both fast and slow process corners and also authored the report detailing these aspects.

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